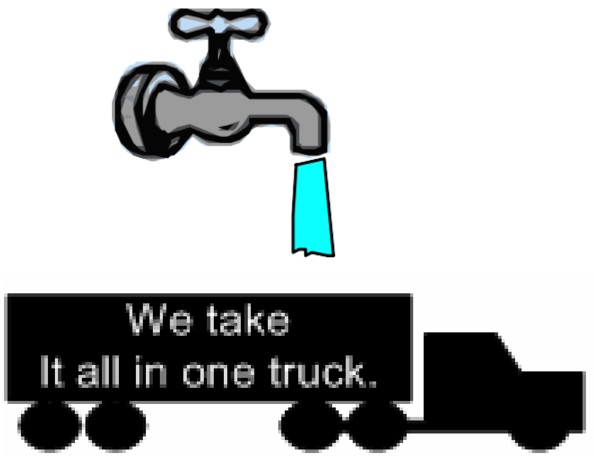
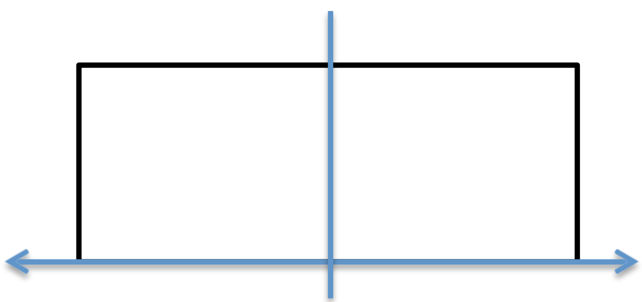
Orthogonal Frequency Division Modulation (OFDM)

Basic Concept of OFDM

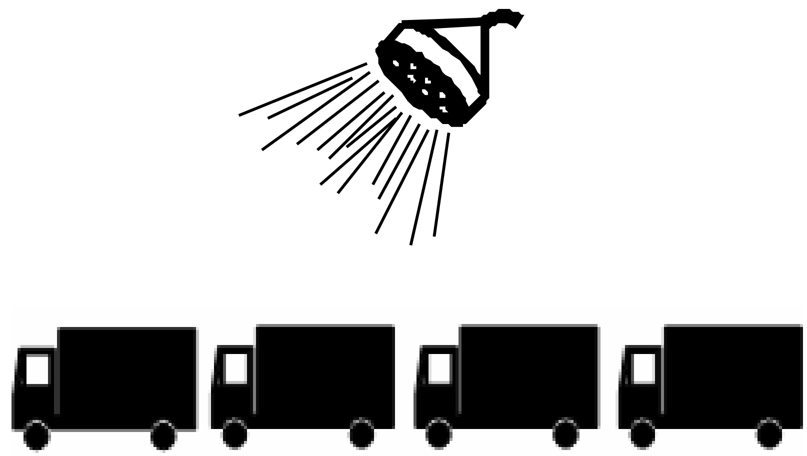
Wide-band channel

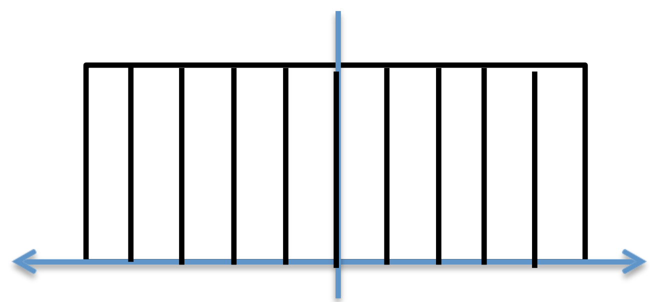




Send a sample using the entire band

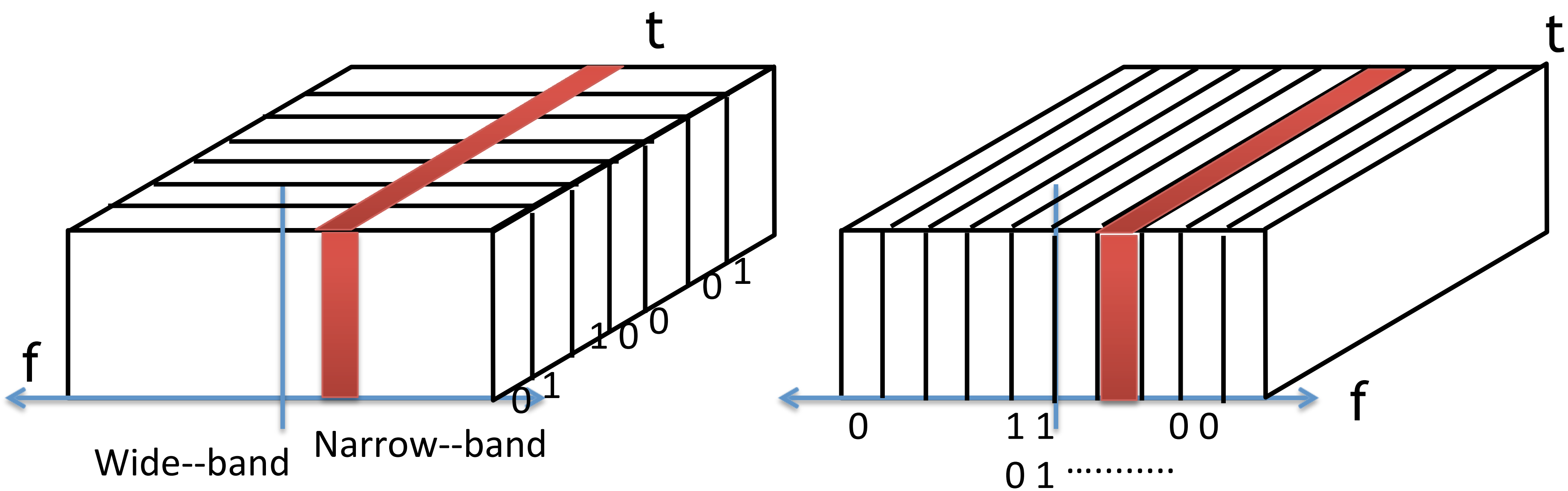
Multiple narrow-band channels





Send samples concurrently using multiple **orthogonal sub-channels**

Why OFDM is better?



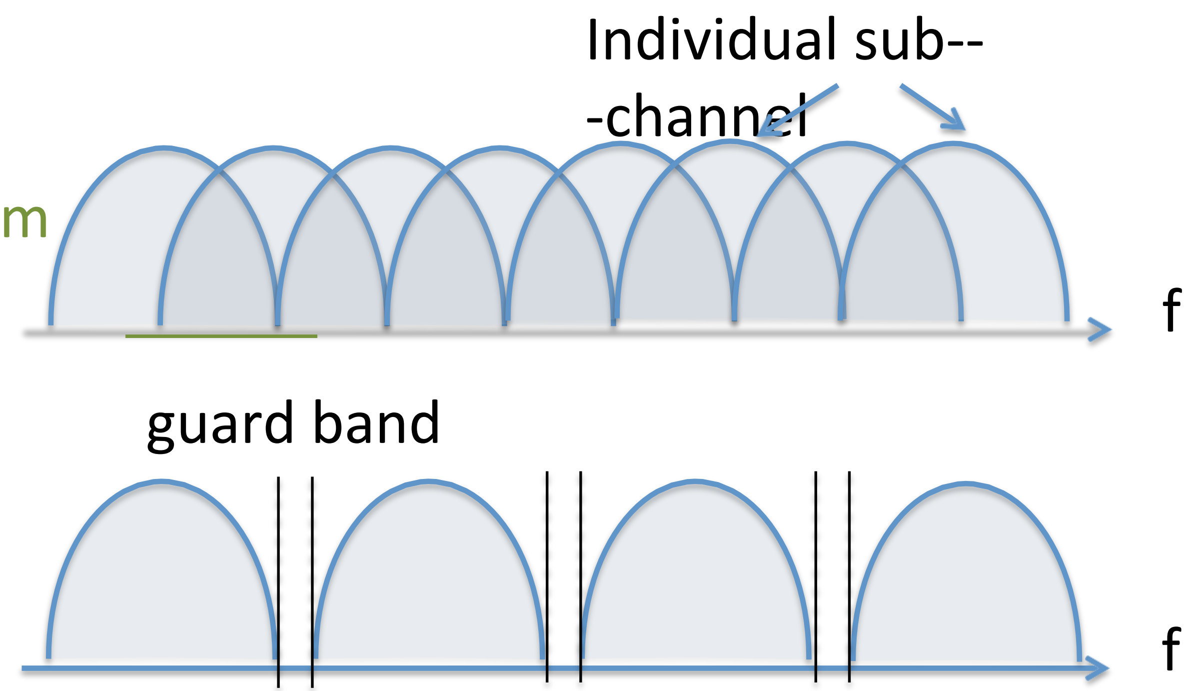
- Multiple sub-channels (sub-carriers) carry samples sent at a lower rate
  - Almost same bandwidth with wide-band channel
- Only some of the sub-channels are affected by interferers or multipath effect

Importance of Orthogonality

- Why not just use FDM (frequency division multiplexing)
  - Not orthogonal

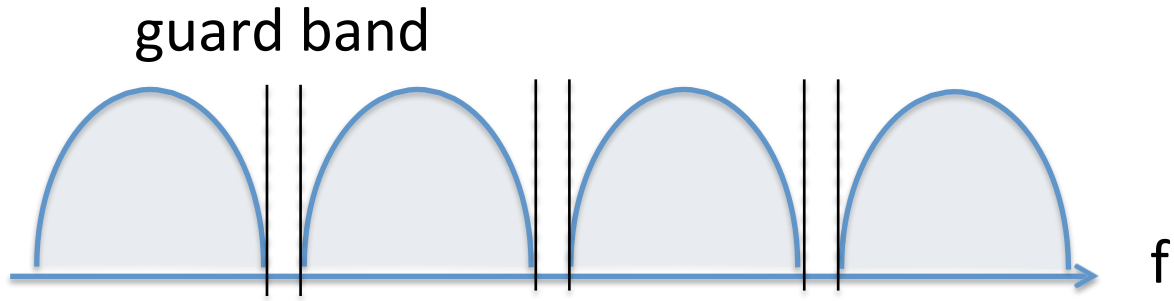
Leakage interference from adjacent sub--channels

Guard bands protect leakage interference

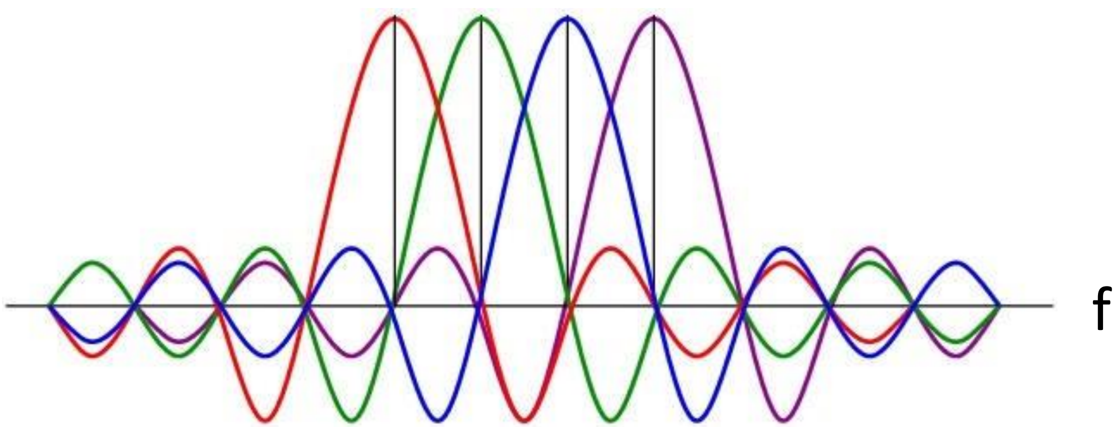


- Need guard bands between adjacent frequency bands extra overhead and lower throughput

Difference between FDM and OFDM



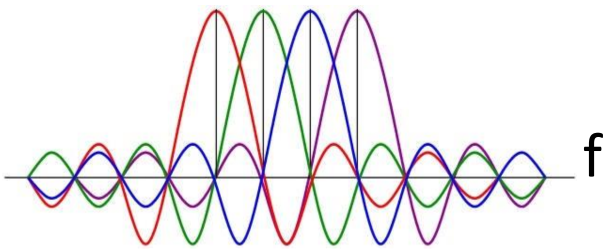
Frequency division multiplexing



Orthogonal sub-carriers in OFDM

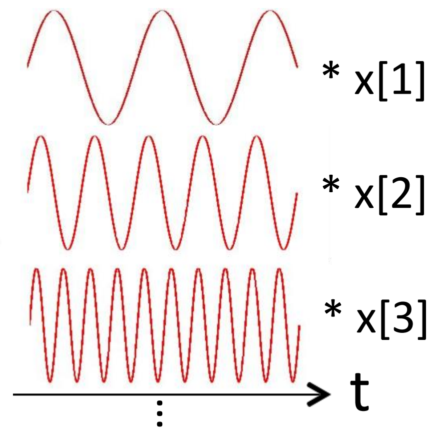
Don’t need guard bands

Orthogonal Frequency Division Modulation



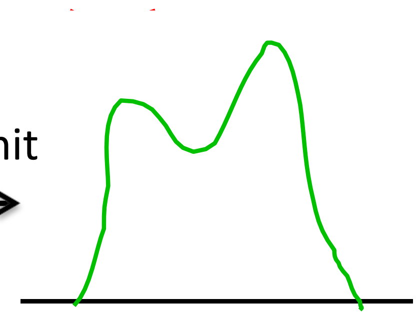
Data coded in frequency domain

IFFT



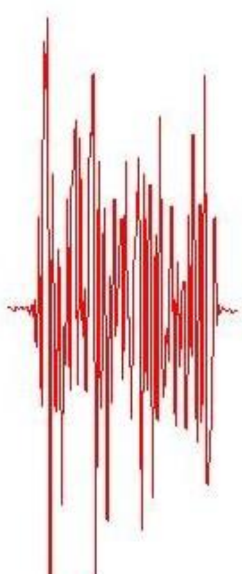
Transformation to time domain:
each frequency is a sine wave
In time, all added up

transmit



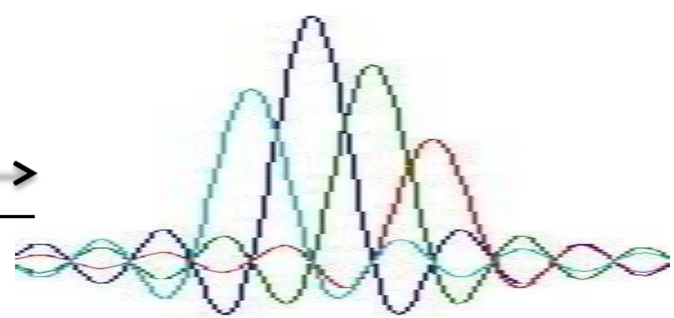
Channel frequency response

receive



Time domain signal

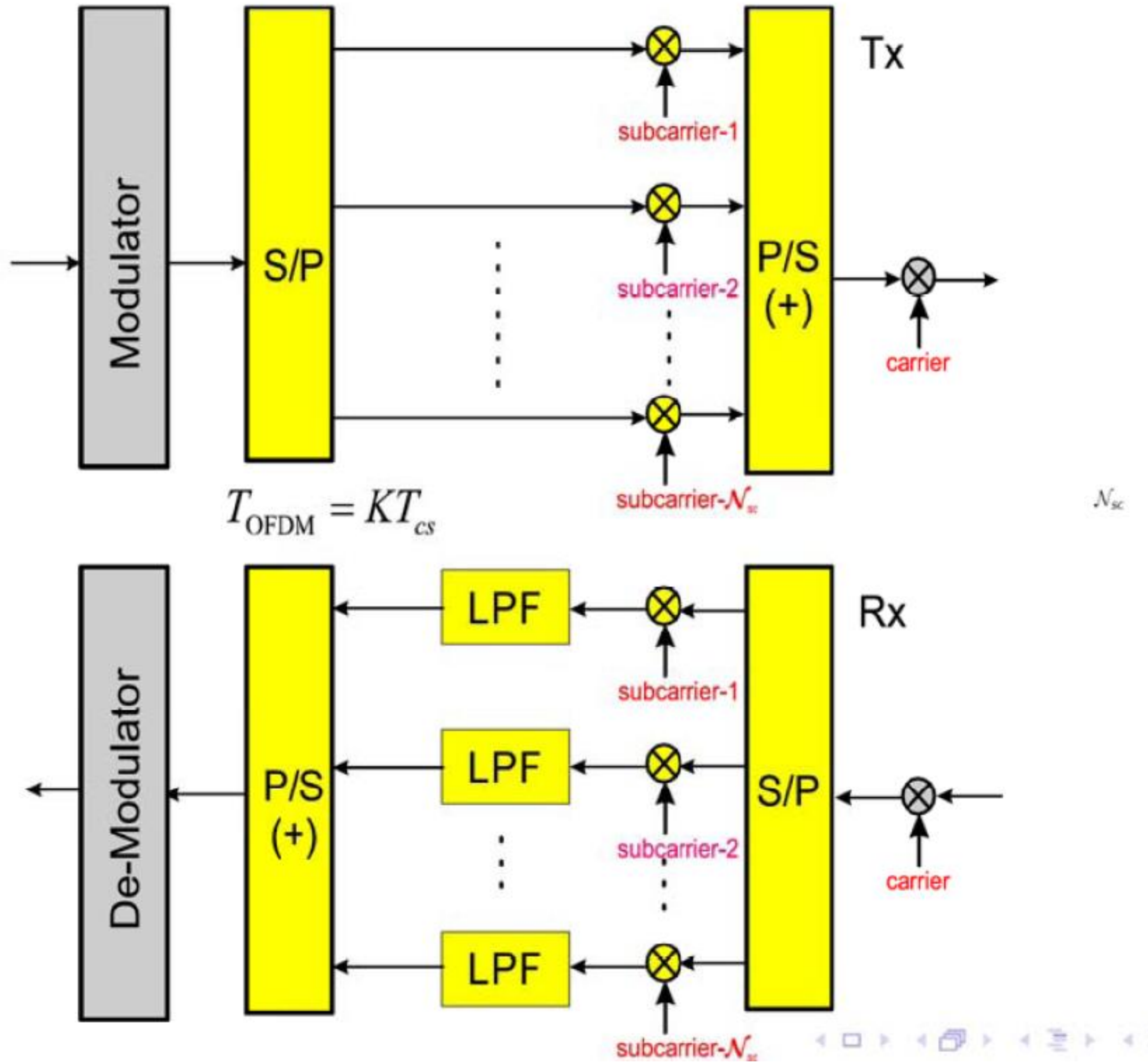
FFT



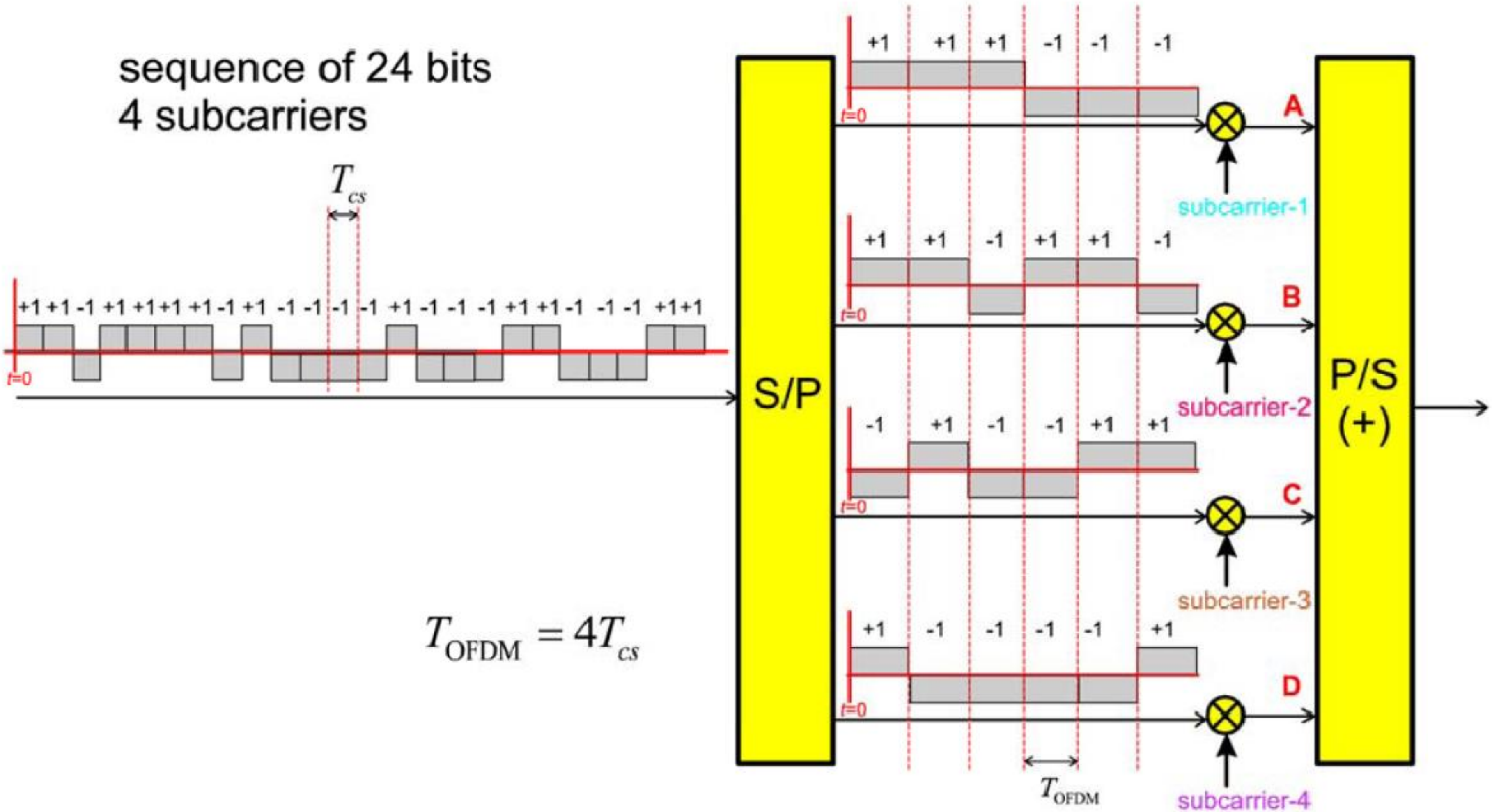
Frequency domain signal

Decode each subcarrier separately

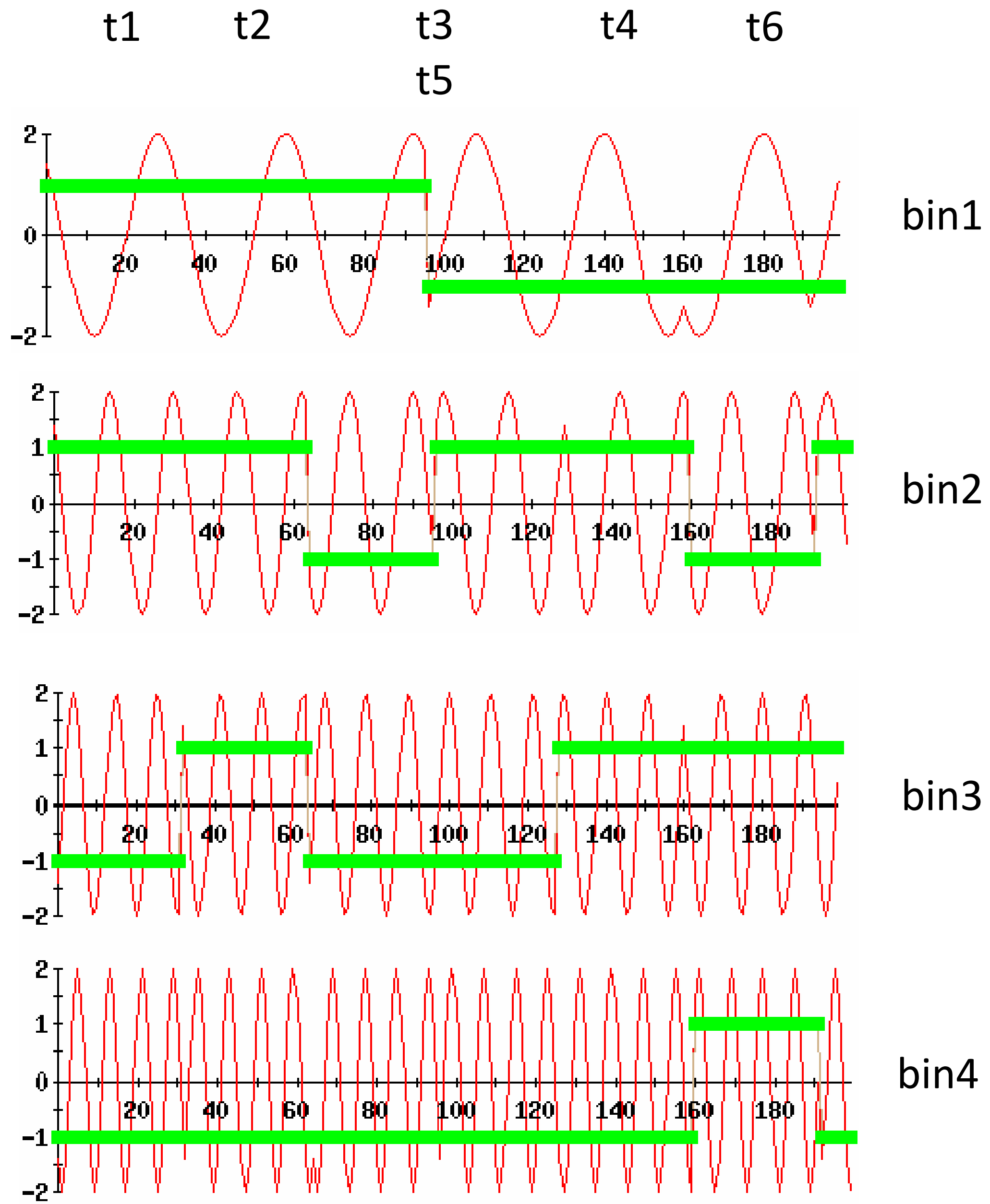
OFDM - Analogue Block Diagram



OFDM TX: Example of 4 Subcarriers

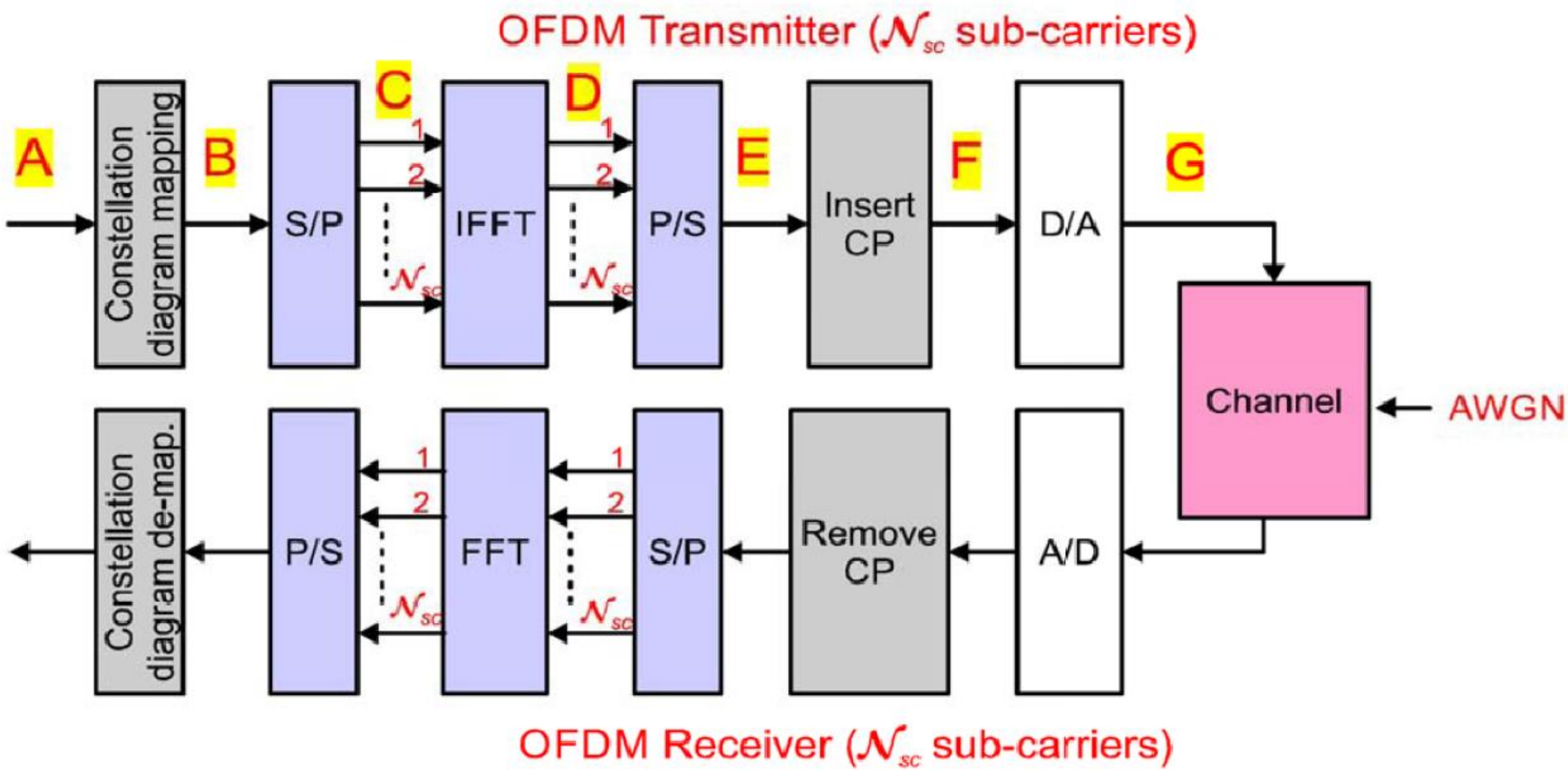


symbol1	1	1	--1	--1
symbol2	1	1	1	--1
symbol3	1	--1	--1	--1
symbol4	--1	1	--1	--1
symbol5	--1	1	1	--1
symbol6	--1	--1	1	1



OFDM Transmitter and

OFDM - Digital Block Diagram (in Practice)

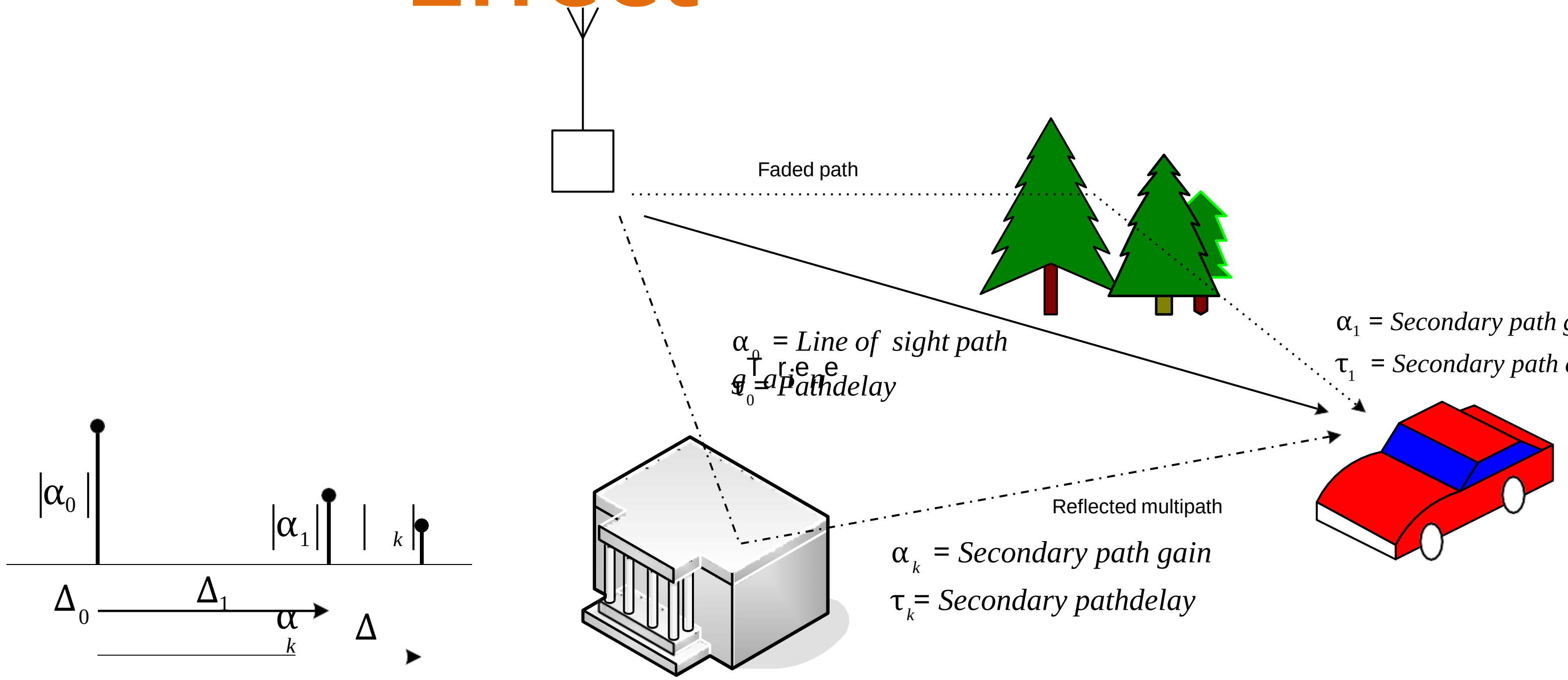


- The stream of modulated symbols is demultiplexed onto $\mathcal{N}_{sc}$ **parallel streams** (using a S/P converter) - where $\mathcal{N}_{sc}$ is the number of subcarriers
- The $\mathcal{N}_{sc}$ parallel streams are fed to the **IFFT** .
- **Remember** :

Frequency-Domain (FD) samples	Time-Domain (TD) Samples
$X[k] \overset{\text{iFFT}}{\longmapsto}$	$x(t_\ell) = \frac{1}{\mathcal{N}_{sc}} \sum_{k=0}^{\mathcal{N}_{sc}\text{-}1} X[k] \exp(j2\pi k \frac{\ell}{\mathcal{N}_{sc}})$
$X[k] = \sum_{\ell=0}^{\mathcal{N}_{sc}\text{-}1} x(t_\ell) \exp(-j2\pi k \frac{\ell}{\mathcal{N}_{sc}})$	$\overset{\text{FFT}}{\longleftarrow} x(t_\ell)$
$\forall k = 0, 1, ..., \mathcal{N}_{sc} - 1$	$\forall \ell = 0, 1, ..., \mathcal{N}_{sc} - 1$

- Orthogonality: $\sum_{\ell=0}^{\mathcal{N}_{sc}-1} \exp(-j(2\pi k \frac{\ell}{\mathcal{N}_{sc}}) \exp(-j(2\pi m \frac{\ell}{\mathcal{N}_{sc}}) = 0, \forall k \neq m$
- The time samples at the output of the iFFT are multiplexed using a **P/S Converter** ,
- **Cyclic prefixes** are inserted.

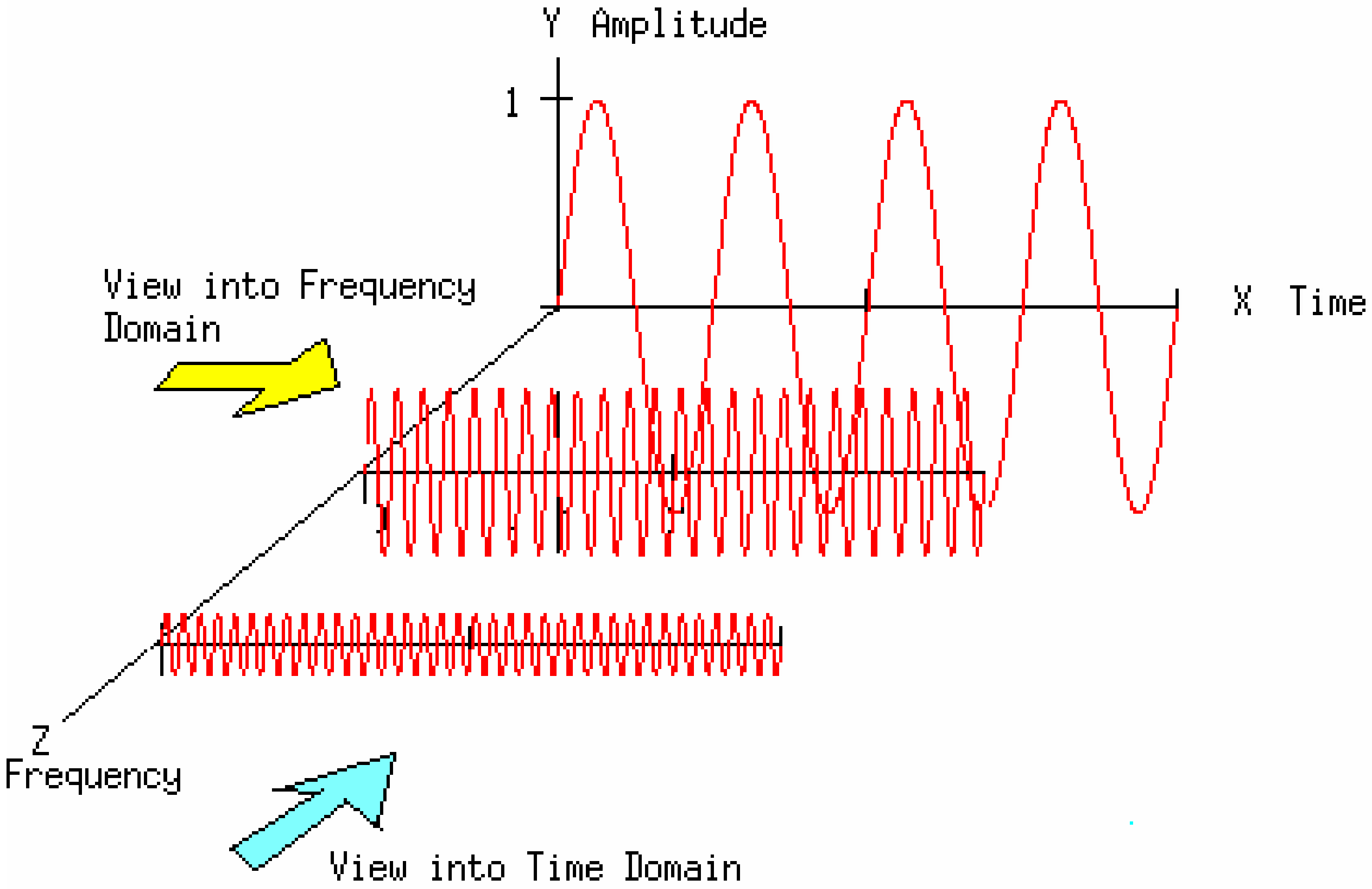
Multi-Path Effect



$$y(t) = h(0)x(t) + h(1)x(t-1) + h(2)x(t-2) + \cdots$$
$$= \sum_{\Delta} h(\Delta)x(t-\Delta) = h(t) \text{©} x(t) \quad \Leftrightarrow \quad Y(f) = H(f)X(f)$$

time-domain

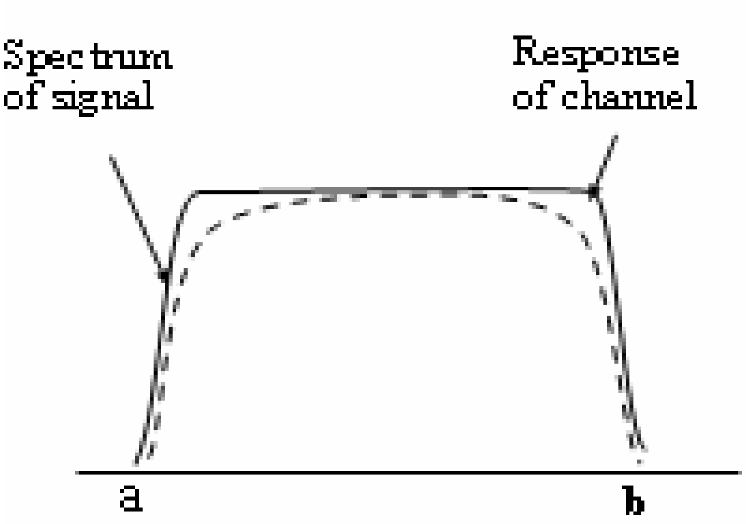
frequency-domain

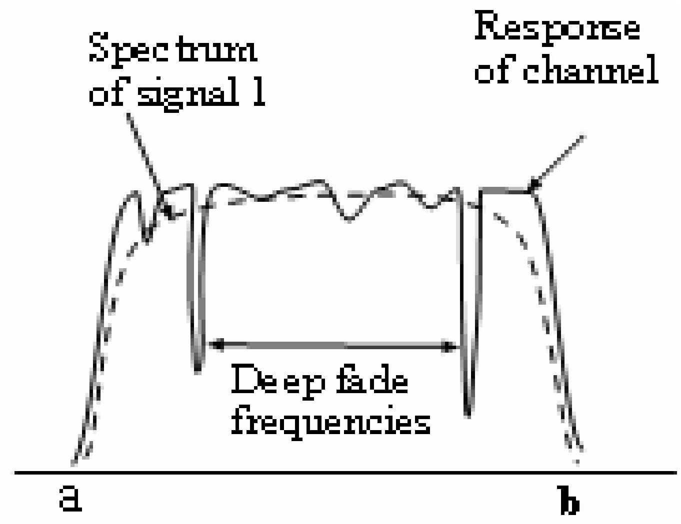


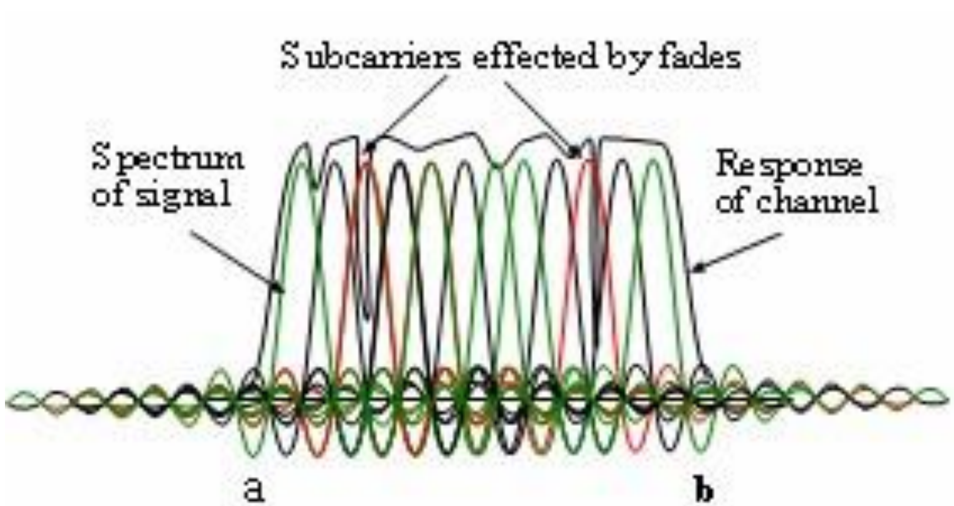
Current symbol + delayed-version symbol

- Signals are deconstructive in only certain frequencies

Frequency Selective



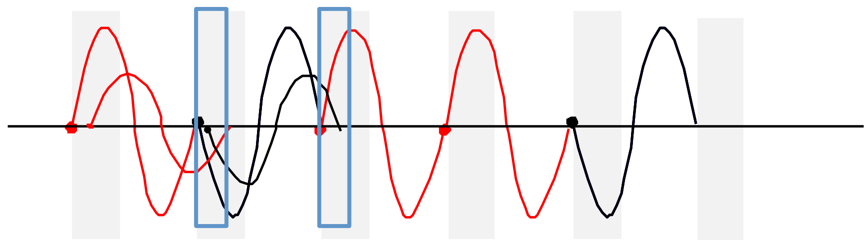




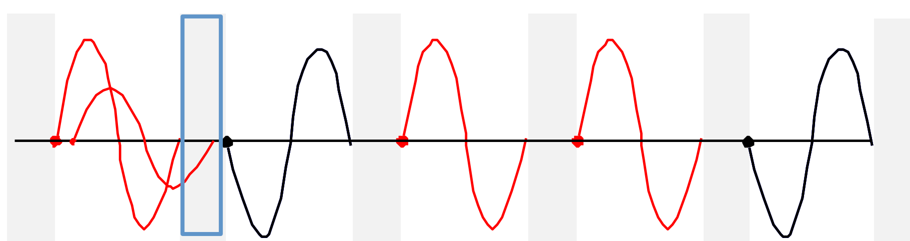
Frequency selective fading: Only some sub--carriers get affected

Inter Symbol Interference (ISI)

- The delayed version of a symbol overlaps with the adjacent symbol



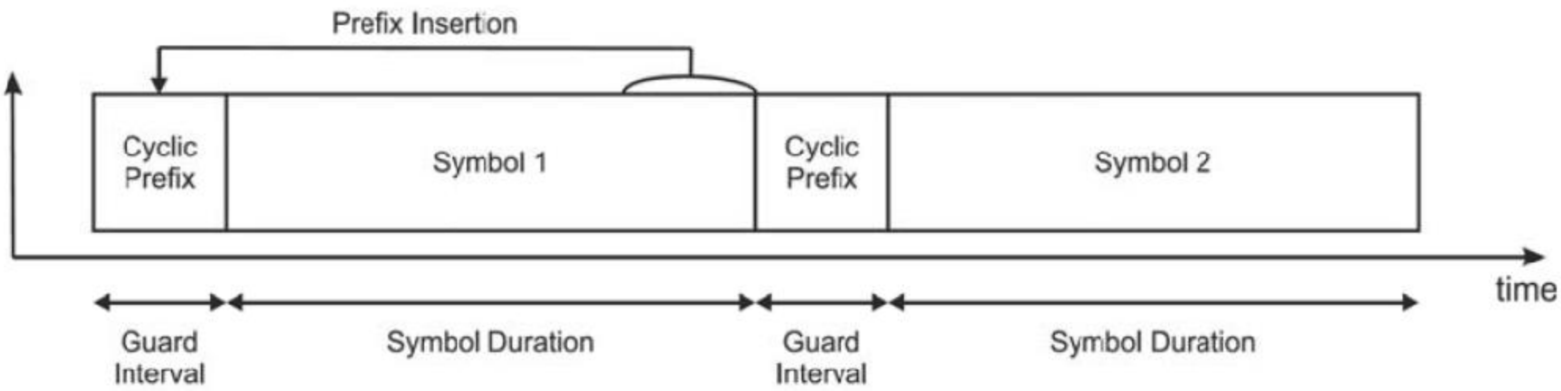
- One simple solution to avoid this is to introduce a guard-band



Guard band

Cyclic Prefix, CP

- The **cyclic prefix** is a **guard interval (GI)** placed to protect OFDM signals from intersymbol interference in the presence of multipath.



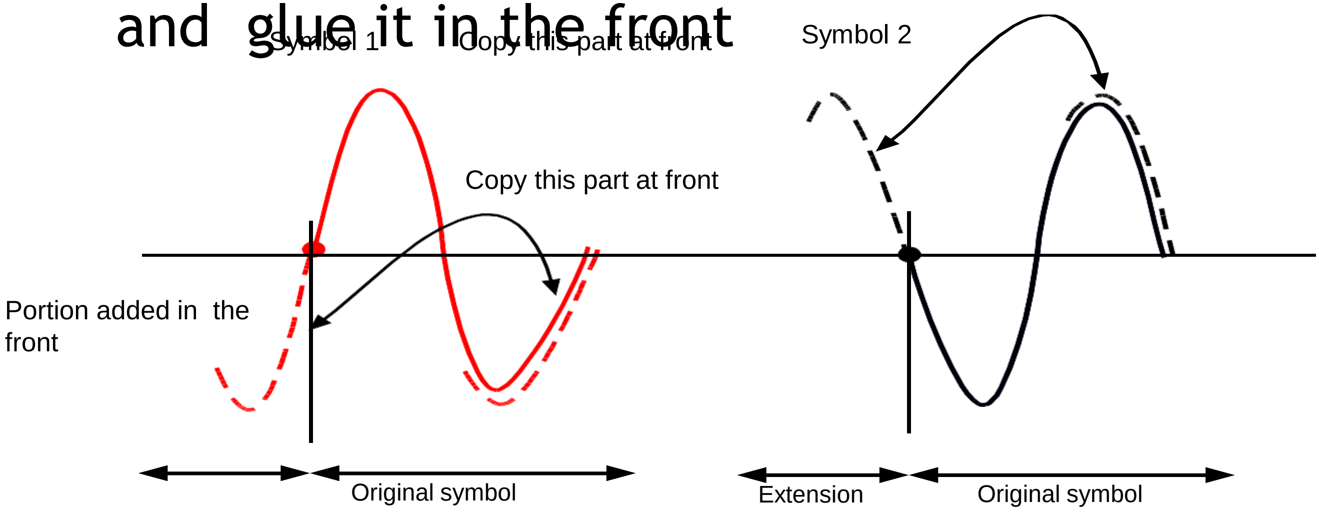
- The last section of a symbol is used as a prefix in the front of the symbol.
- The duration of the guard interval T_{GI} should be

$$T_{\mathrm{GI}} > T_{\mathrm{spread}}, \tag{1}$$

to minimise **inter-symbol interference** (ISI).

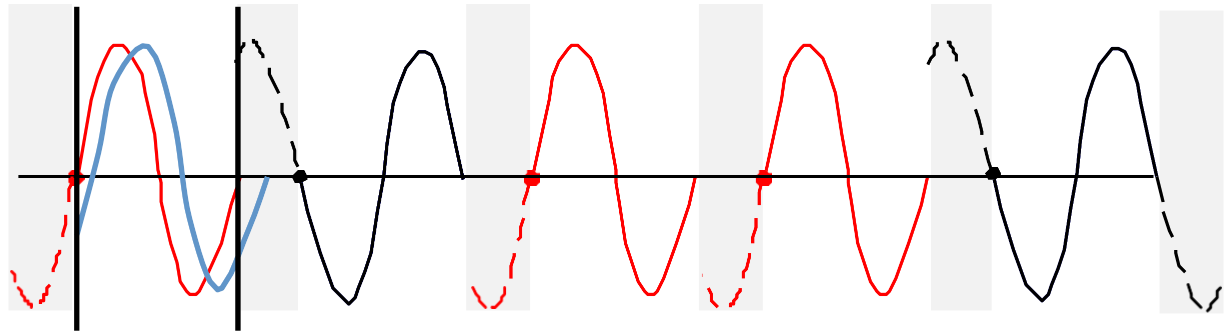
Cyclic Prefix (CP)

- However, we don't know the delay spread exactly
  - The hardware doesn't allow blank space because it needs to send out signals continuously
- Solution: Cyclic Prefix
  - Make the symbol period longer by copying the tail and glue it in the front



In 802.11,
CP:data = 1:4

Cyclic Prefix (CP)



- Because of the usage of FFT, the signal is periodic

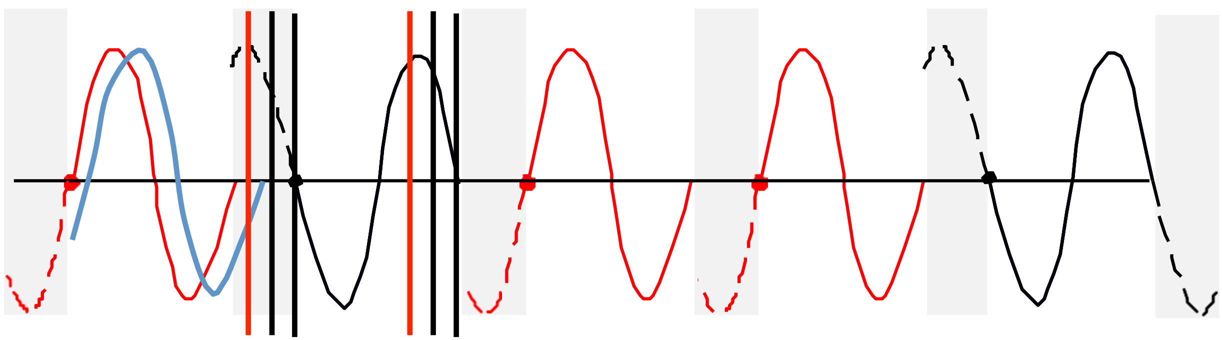
$$\text{FFT}(\quad) = \exp(--2j\pi_{\Delta}f)*\text{FFT}(\quad)$$

delayed version original signal

- Delay in the time domain corresponds to rotation in the frequency domain
  - Can still obtain the correct signal in the frequency domain by compensating this rotation

Side Benefit of CP

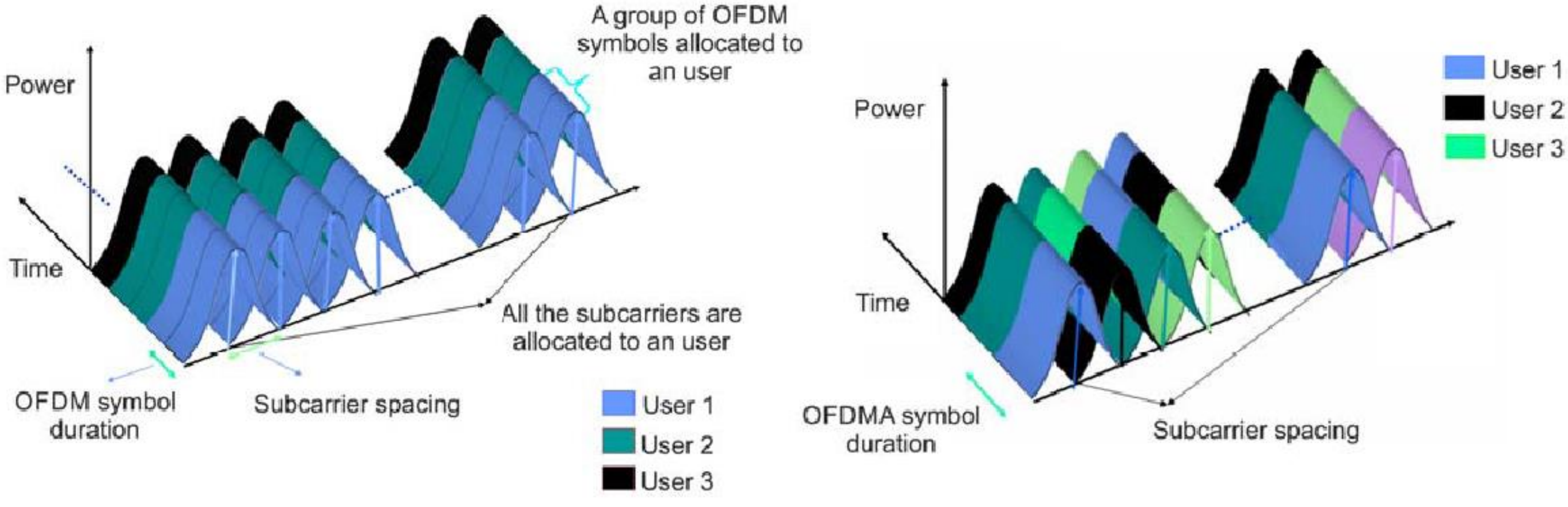
- Allow the signal to be decoded even if the packet is detected after some delay



undecodable

OFDM vs OFDMA

- **OFDMA** is the "**access**" version of OFDM, that is a scheme where the access to resources is shared between users.



(a) OFDM-TDMA Signal (B) OFDMA Signal

- In **OFDM-TDMA**, time-slots are attributed to each users and the whole OFDM symbol is dedicated to one user.
- In **OFDMA**, both time and frequency are distributed among users.

Benefits Or Advantages Of OFDMA

➡Unlike OFDM which allocates all the subcarriers of the OFDM symbol to one user, OFDMA allocates a subset of subcarriers to different users. Hence OFDMA makes efficient use of frequency allocations in comparison to FDMA and OFDM.
➡It provides multi-user diversity. This is because OFDMA allows different users to transmit over different parts of the frequency spectrum known as traffic channels. Hence different users see different channel qualities. Hence deep faded channel for user#1 may still be useful for the other users.
➡It makes the receiver simple as it eliminates intra-cell interference. Here only FFT processing is needed.
➡Better performance in a Fading environment.
➡Increased data rate as the carriers are orthogonal, the adjacent spectrum can overlap.
➡Reduces Inter Symbol Interference(ISI) using Cyclic Prefix

Drawbacks Or Disadvantages Of OFDMA

➡The permutation and determination rules of subcarriers for allocation and deallocation to subchannels are complex. This makes transmitter and receiver algorithms complex for data processing/extraction, unlike OFDM.

➡OFDMA has a higher PAPR (Peak to Average Power Ratio). Hence large amplitude variations lead to an increase of in-band noise.

➡High cost compared to other techniques.

Applications of OFDMA

OFDMA technology can be applied anywhere data is sent along radio waves, including the:

- Mobility mode of the IEEE 802.16 wireless standard known as WiMAX
- Wireless LAN (WLAN) standard IEEE 802.11ax (Wi-Fi 6)
- IEEE 802.20 mobile wireless metropolitan-area network (WMAN) standard
- Downlink of the 3GPP Long-Term Evolution (LTE) fourth-generation mobile broadband standard (4G)
- OFDMA is used in the air interface stage of 5G New Radio